



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Strategic insights into spending, segments & subscriptions

Dataset at a Glance

3,900

Transactions

18

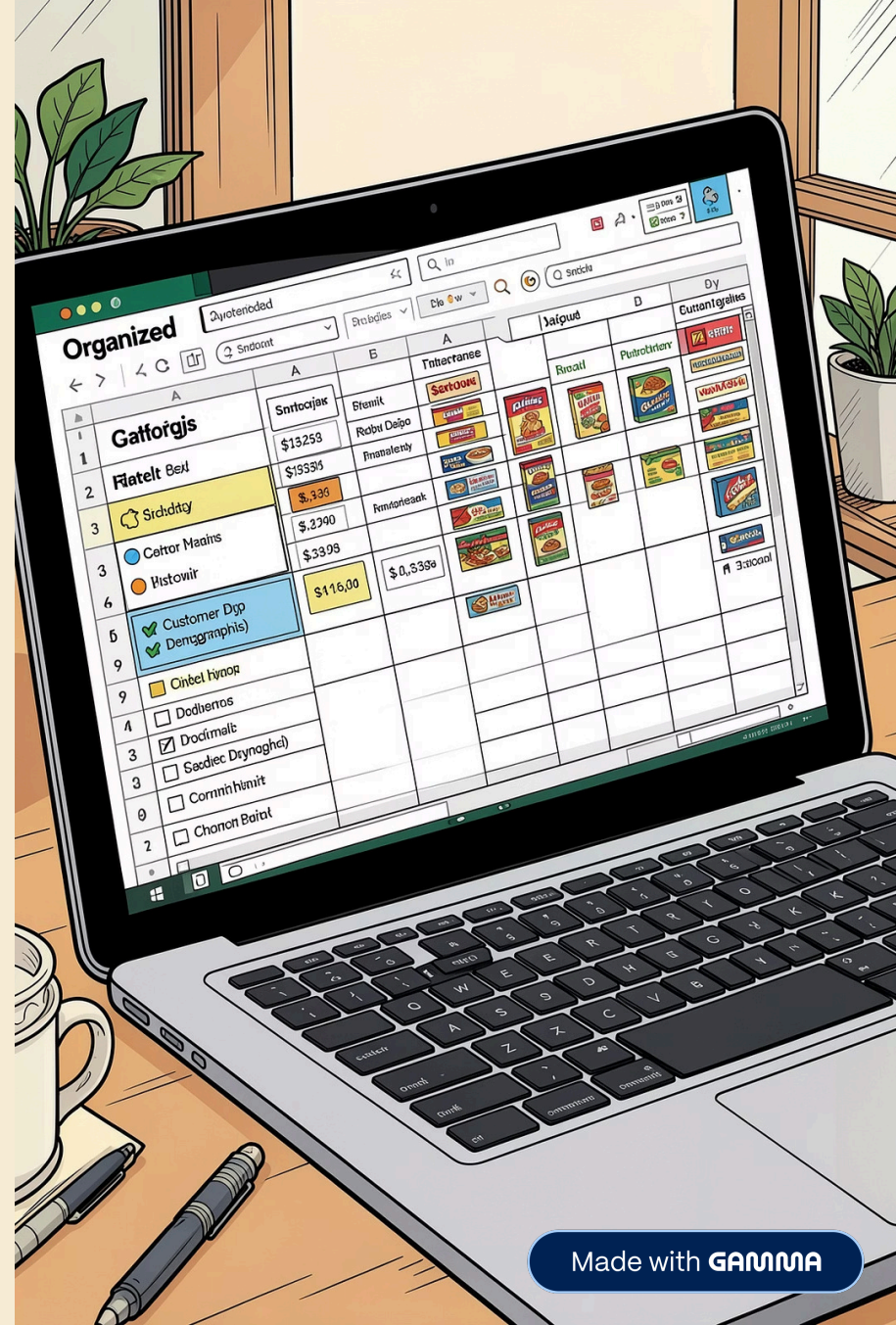
Features

37

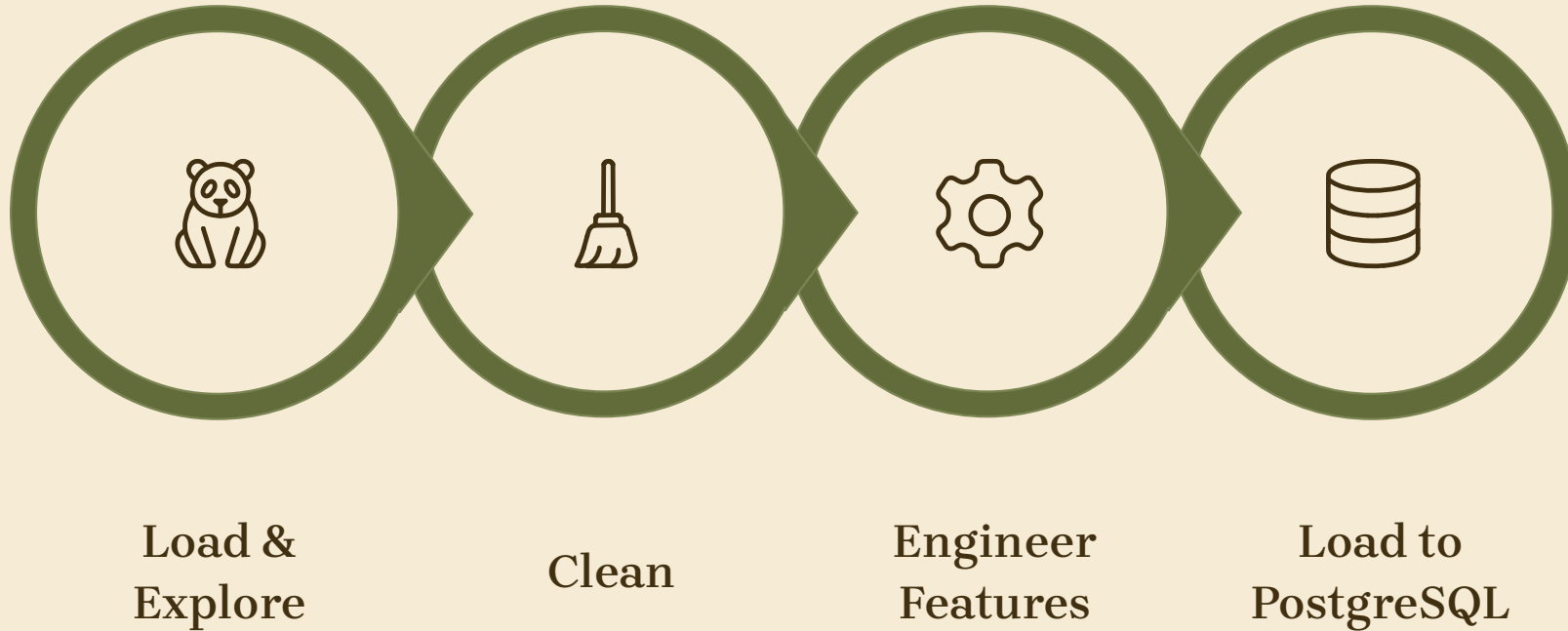
Missing Values

Review Rating column

Demographics, purchase details, behavior signals — age, gender, category, discount, subscription & more

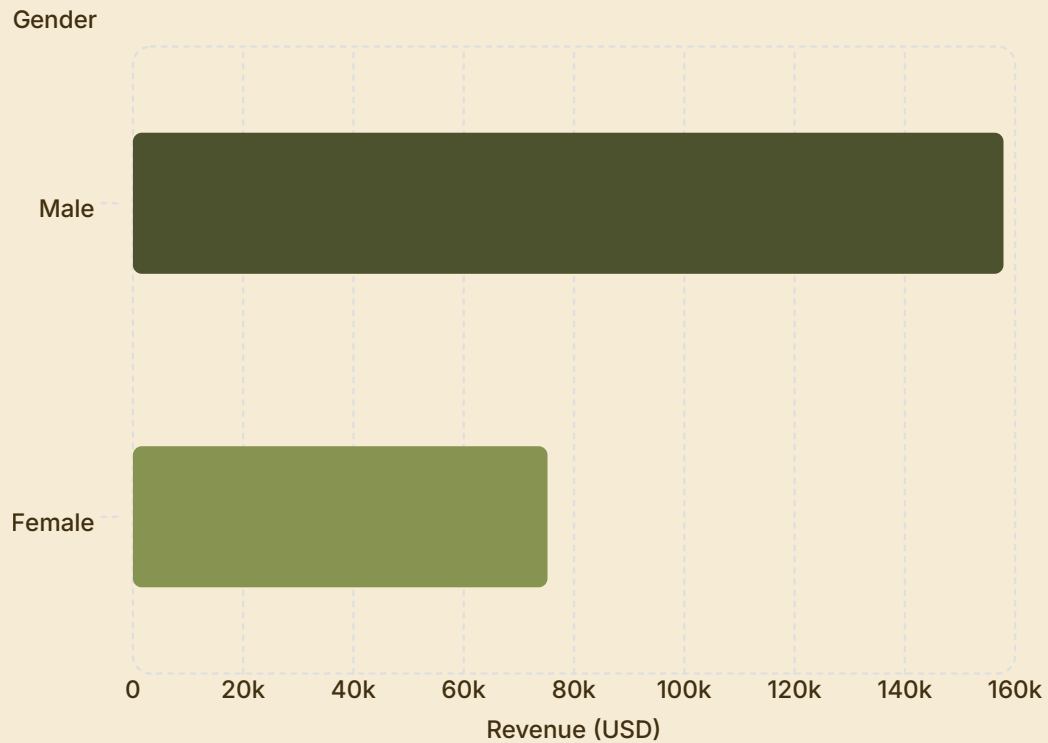


Data Preparation in Python



Columns standardized to snake_case; promo_code_used dropped as redundant with discount_applied

Revenue by Gender



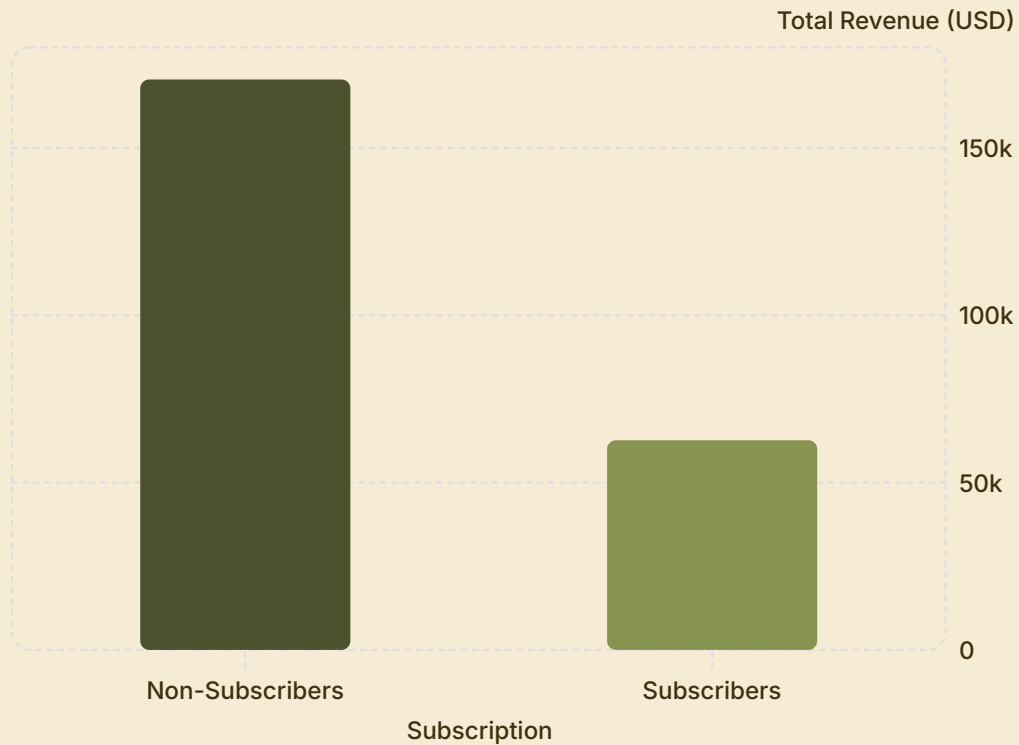
Male customers drive 2× more revenue

Male: **\$157,890**

Female: **\$75,191**

2,652 of 3,900 purchases were made by male customers

Subscribers vs. Non-Subscribers



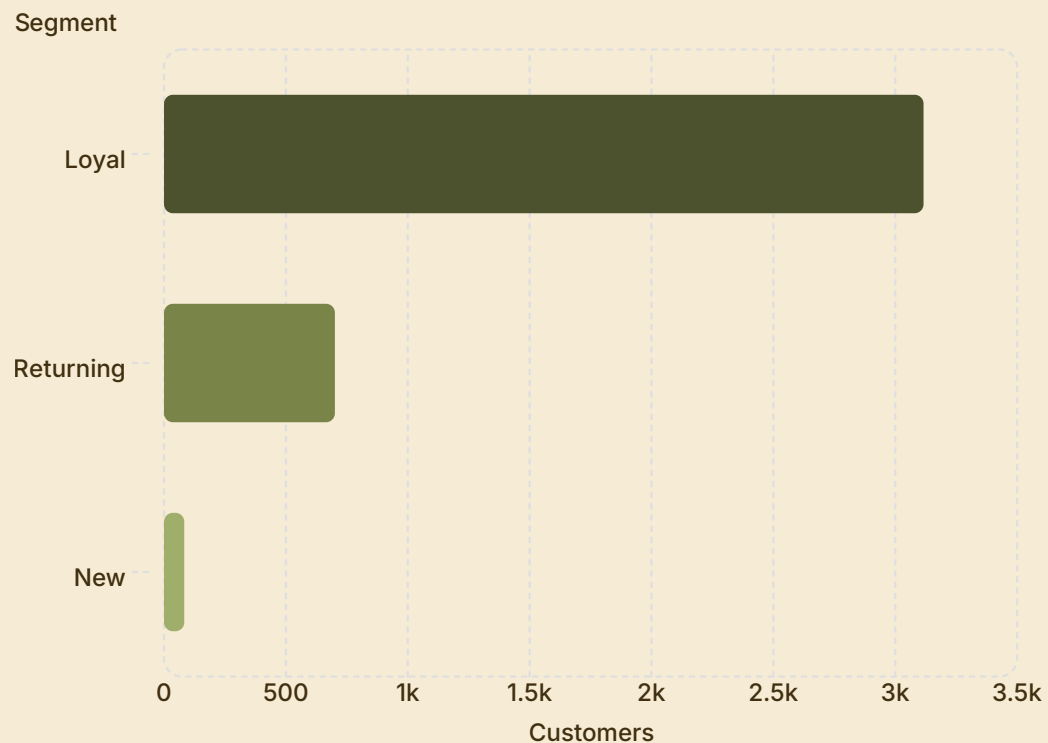
Non-subscribers dominate volume

73% non-subscribers · 2,847 customers

27% subscribers · 1,053 customers

Average spend nearly identical (~\$59.50)

Customer Segmentation

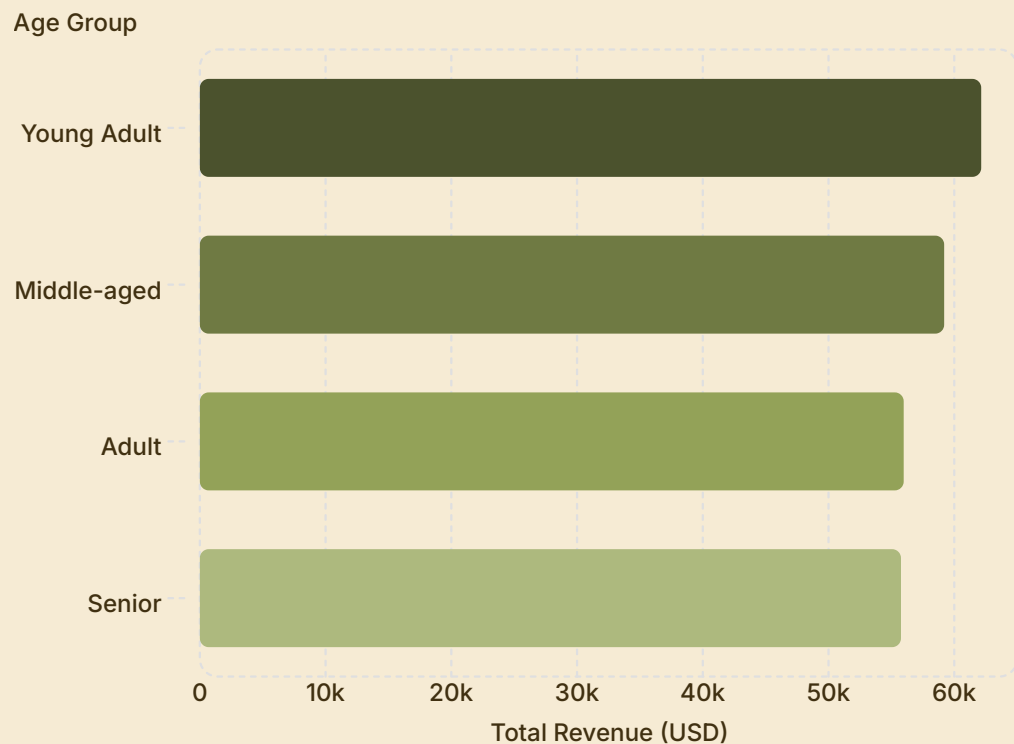


Loyal customers dominate the base

3,116 Loyal · 701 Returning · 83 New

958 of 1,379 repeat buyers (5+ purchases) hold subscriptions

Revenue by Age Group



Young Adults lead revenue

Ages 18–70 · mean age 44

Revenue spread is relatively even across all groups

Young Adults: **\$62,143** · Seniors: **\$55,763**

Top Products & Discount Patterns

Highest-Rated Products

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%



Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

Filters

Subscription, Gender, Category, Shipping



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users